

# RegDivergence-101: An LLM Benchmark for Cross-Jurisdiction Regulatory Contradiction Detection in Life Sciences

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Pharmaceutical sponsors developing a drug for both the United States and the European Union must reconcile guidance issued independently by the FDA and the EMA. Where the two agencies require *substantively the same thing*, a sponsor can file once; where they *diverge*, a single trial design risks rejection in one region; where one agency is *silent* on a point the other regulates, the sponsor must infer obligations. Today this reconciliation is performed manually by regulatory-affairs experts. We introduce **cross-jurisdiction regulatory divergence detection**: given an FDA requirement and an EMA requirement on the same topic, classify their relationship as AGREE, DIVERGE, or SILENT. SILENT is inherently directional (SILENT\_FDA vs. SILENT\_EMA); we record direction per pair and report per-direction F1 alongside the collapsed label. We release **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark (labels grounded in three peer-reviewed FDA/EMA comparison studies and primary FDA/EMA/ICH guidance text; dual-annotation inter-annotator  $\kappa = 0.85$ ), and systematically characterise a four-method baseline hierarchy: lexical heuristic (0.511 macro-F1, 95% CI [0.411–0.605]), NLI cross-encoder (0.233), obligation-level Graph-RAG (0.663 [0.570–0.747]), and flat LLM judge / Claude Haiku (0.830 [0.747–0.908]). Three directional observations emerge at pilot scale ( $n = 101$ ): SILENT is semantically detectable but invisible to entailment-only formulations; pair-level obligation graphs improve over lexical methods but trail flat-LLM context (CIs partially overlapping); and corpus-level graph construction is the indicated architectural target for large-scale silent-detection. RegDivergence-101 is a pilot release establishing the task formulation and baseline hierarchy; four unrepresented regulatory domains and an expansion roadmap are described in §7.

CCS Concepts: • **Computing methodologies** → **Lexical semantics**; • **Applied computing** → **Health informatics**.

Additional Key Words and Phrases: large language models, LLM benchmark, regulatory NLP, AI for compliance, FDA/EMA divergence, contradiction detection, Graph-RAG, retrieval-augmented generation, generative AI, life sciences, AGREE/DIVERGE/SILENT classification

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## 1 Introduction

Bringing a medicine to market in both the US and EU requires satisfying two regulators that publish guidance separately and frequently *differently*. Gene-therapy sponsors face a 15-year long-term follow-up recommendation from FDA while EMA defers the duration to case-by-case risk assessment—same topic, materially different specificity. Analogous divergences span pediatric extrapolation, endpoint definitions in ulcerative colitis, biosimilar comparability, and estimand implementation. A sponsor who misses one divergence during protocol design can lose a year or more re-running a study.

Reconciling the two corpora is currently a manual, expert task. Yet the NLP building blocks for automating it all exist—document-level NLI (ContractNLI [1], DocNLI [2]), legal contradiction detection (LegalWiz [3], CLAUSE [4]), and Graph-RAG (Microsoft GraphRAG [5], GraphCompliance [6], RAGulating Compliance [7]). **No prior work combines them to detect divergence between two parallel regulatory corpora from different jurisdictions.** All existing contradiction work is *intra-document* (within one contract) or *single-jurisdiction* (contract-vs-statute).

This paper makes three contributions: (1) **Task formulation** — three-way classification (AGREE / DIVERGE / SILENT) over pre-aligned FDA/EMA requirement pairs; SILENT treated as directional (SILENT\_FDA / SILENT\_EMA) with per-direction F1 in Table 2. (2) **Pilot benchmark** — **RegDivergence-101**, 101 pairs across 13 topic areas with a documented two-track construction protocol,  $\kappa = 0.85$ , and bootstrap CIs; expansion roadmap in §7. (3) **Baseline hierarchy** — four methods compared at pilot scale ( $n = 101$ ) with bootstrap 95% CIs, yielding three directional observations on the task’s structure.

## 2 Related Work

**Document-level NLI.** ContractNLI [1] introduced document-level entailment/contradiction/not-mentioned classification with evidence extraction, and showed that contradiction detection lags entailment, especially under “negation by exception.” DocNLI [2] extended NLI to full documents.

**Legal contradiction detection (2025–2026).** LegalWiz [3] uses a multi-agent framework for legal contradiction detection but is confined to single-document regimes; Better Call CLAUSE [4] benchmarks 7,500+ perturbed contracts and finds LLMs struggle to justify subtle errors legally. A published RAG-robustness study directly informs our results: Yoran et al. [8] show that retrieved context which is irrelevant or non-entailing can mislead a retrieval-augmented model unless it is explicitly filtered—the failure mode in Observation 2, where modal-register similarity misleads the pair-level graph.

**Graph-RAG and regulatory compliance.** Microsoft GraphRAG [5] builds entity graphs with community summaries for local and global queries. GraphCompliance [6] aligns *policy graphs* (regulatory requirements) with *context graphs* (organisational facts) for GDPR; it explicitly lists “handling conflicting

regulatory interpretations” as open. RAGulating Compliance [7] builds an ontology-free regulatory knowledge graph for traceable QA.

**Cross-jurisdiction regulatory benchmarks.** The closest concurrent work is **Sino-US-DrugQA** [9] (11,871 MCQA items, US–CN), which shares our concept-alignment thesis but cannot express the SILENT class. Comparative-regulatory-science surveys document FDA/EMA divergence across dosing [10], oncology labelling [11], and multi-country SmPCs [12], but as human-authored comparisons, not ML benchmarks. RegDivergence-101 is the first FDA/EMA benchmark with an explicit SILENT class. In high-stakes sectors such as healthcare, policy-aligned evaluation frameworks weigh safety and regulatory constraints over raw accuracy [13]; RegDivergence-101 extends this to cross-jurisdiction regulatory reasoning.

**Gap.** Contradiction detection is intra-document or single-jurisdiction; Graph-RAG targets single-corpus QA or single-jurisdiction compliance; FDA/EMA divergence is documented manually but never automated. Our task sits precisely in this gap.

### 3 Task and Dataset

#### 3.1 Task Definition

For a topic  $t$ , let  $f_t$  be an FDA requirement statement and  $e_t$  the corresponding EMA statement. The label  $y_t$  is:

- **AGREE** — both jurisdictions require substantively the same thing (same modal strength, same threshold, same scope of applicability). Paraphrase equivalence and synonymous modals (*shall* / *must*) are treated as AGREE; minor drafting differences that do not alter sponsor obligations are disregarded.
- **DIVERGE** — the requirements directly conflict: they impose different thresholds, different modal strengths (*shall* vs. *should*), different timelines, or mutually incompatible design specifications.
- **SILENT** — one jurisdiction does not address what the other regulates. SILENT is *directional*: **SILENT\_FDA** (FDA silent, EMA regulates) vs. **SILENT\_EMA** (EMA silent, FDA regulates). Direction is recorded in the `silent_direction` field and reported in Table 2; a unified SILENT label is retained in Table 1 for backward compatibility.

**Epistemic status of SILENT labels.** A SILENT label is an absence claim: it asserts that no counterpart requirement was found after an exhaustive corpus search. This carries irreducible epistemic uncertainty. We bound this uncertainty through (a) a documented corpus-search protocol (§3.3) and (b) a per-pair `doc_confidence` flag (HIGH / MODERATE). Downstream work should treat MODERATE-confidence SILENT pairs as soft labels.

**Task scope.** This formulation assumes requirement pairs are *pre-aligned*—both sides address the same regulatory topic, established via the two-track protocol (§3.2). Topic alignment is a non-trivial upstream problem treated separately; the corpus-level architecture that would subsume it is described in §7.

### 3.2 Pair Construction Protocol

RegDivergence-101 is constructed via two tracks that differ in how topical correspondence between FDA and EMA requirements is established. Per-pair source-track attribution (**track**: A or B) is released with the dataset.

**Track A — expert-comparison-derived pairs (59 pairs).** Extracted from three open-access peer-reviewed studies that comparatively mapped FDA and EMA requirements: Vieujean et al. [14] (ulcerative colitis, 40 pairs), Boesen et al. [15] (psychiatric drug trials, 10 pairs), and Schwarz and Decristoforo [16] (radiopharmaceuticals, 9 pairs). Domain experts in those studies had already established co-topicality; our task was to classify the AGREE/DIVERGE/SILENT relationship between their pre-aligned pairs.

**Track B — primary guidance pairs (42 pairs).** For pairs from ICH E9(R1), E11A [17], and FDA/EMA biosimilar and gene-therapy guidance, co-topicality was established by (1) anchoring requirements to a shared ICH Common Technical Document (CTD) module/sub-topic, then (2) expert confirmation that the candidate pair was genuinely co-topical.

**Specificity exclusion criterion.** Candidate pairs operating at different regulatory hierarchy levels (e.g., *statistical analysis plan* vs. *general statistical principles*) were excluded to prevent conflating level-of-detail differences with substantive DIVERGE labels. Excluded pairs and the full exclusion-criteria checklist are released in `excluded_pairs.jsonl` for third-party audit.

Each pair records `{id, topic, track, source, fda_text, ema_text, label, silent_direction, doc_confidence, disputed, annotation_notes}`. The combined release contains **101 pairs**: 42 AGREE, 38 DIVERGE, 21 SILENT, across 13 topic areas. We report macro-F1 to weight all classes equally.

### 3.3 Annotation Protocol

**Annotators.** Annotator A (C. Wu) is a solution architect for a regulatory-document AI platform; Annotator B (Z. Zhou) is a product manager for an AI-powered clinical-trial management platform. Neither holds formal RA credentialing; Track A expert-panel papers serve as the external human-expert anchor.

**Stages.** Stage 1: Wu assigned labels via definitions-based framing; Stage 2: Zhou independently re-labelled all 101 pairs under a sponsor-obligation framing to probe framing-independent stability. The  $\kappa$  is a cross-framing consistency measure, not identical-instruction IAA. An independent third-annotator pass is planned for RegDivergence-500.

**IAA.** Human cross-framing  $\kappa = 0.85$  (101 pairs, observed 0.90)—*substantial* per Landis and Koch [18], on par with MedNLI (0.65) and ContractNLI ( $\approx 0.72$ ). SILENT near-perfect ( $\kappa = 0.98$ ); AGREE/DIVERGE more variable ( $\approx 0.79$ – $0.81$ ). A secondary LLM consistency check (36-pair sample) yielded  $\kappa = 0.542$  (test-retest, not human IAA).

**Disagreements (n = 10).** Six resolved by source-paper language; four genuinely borderline pairs retain the Stage 1 label and are flagged `disputed = True` in the dataset with written justifications.

**SILENT confidence.** All 21 SILENT pairs carry `doc_confidence = MODERATE`: exhaustive corpus search was performed (full FDA and EMA guidance, Q&As, reflection papers) but absence claims remain epistemically soft. The HIGH/MODERATE audit is deferred to RegDivergence-500.

## 4 Baselines

### 4.1 Evaluation Protocol

RegDivergence-101 is an **evaluation benchmark**, not a training dataset. All baselines are zero-shot or few-shot; results are reported as a ranking trend ( $\Delta = 0.15$ – $0.60$  macro-F1 between adjacent methods, CIs partially overlapping), not pairwise significance.

All methods use **stratified 5-fold splits** (5 seeds) with **bootstrap 95% CI** (1,000 resamples). Lexical and NLI methods use **nested CV** (inner folds tune thresholds); Graph-RAG and LLM judge have no tunable hyperparameters (LLM judge scored on all 101 pairs, bootstrap CI only).

### 4.2 Lexical Heuristic

No neural model, no API—fully offline and deterministic. Combines TF-IDF cosine similarity with five conflict signals: negation asymmetry, divergence cue pairs (e.g., *single-arm* vs. *randomised*), modal-strength asymmetry (*must/shall* vs. *may/encouraged*), numeric-threshold mismatch, and one-sided specificity (concrete number vs. case-by-case deferral). A shared ICH citation is a strong AGREE signal.

### 4.3 NLI Cross-Encoder

`typeform/distilbert-base-uncased-mnli`, run in both directions; labels derived from contradiction/neutral/entailment probabilities with CV-tuned thresholds. Any NLI model lacking an explicit SILENT class maps absence-of-regulation to *neutral*, structurally collapsing SILENT recall (Observation 1 in §6); stronger encoders improve AGREE/DIVERGE F1 but cannot resolve this gap without task-specific supervision.

### 4.4 Graph-RAG Pair-Level Classifier

For each text, Claude Haiku extracts a policy triplet  $\langle \text{subject}, \text{obligation\_level}, \text{requirement}, \text{conditions} \rangle$  where  $\text{obligation\_level} \in \{\text{MANDATORY}, \text{PROHIBITED}, \text{RECOMMENDED}, \text{PERMITTED}, \text{SILENT}\}$ . Only  $\text{MANDATORY} \leftrightarrow \text{PROHIBITED}$  is a hard-coded DIVERGE rule; all other pairs are escalated to a second LLM call that reasons over the *obligation structure* of the aligned node pair, with explicit calibration that modal-register differences (RECOMMENDED vs. MANDATORY) do not automatically imply DIVERGE.

**Within-model scope.** Both the Graph-RAG classifier and the flat LLM judge use `claude-haiku-4-5-20251001`. The comparison between them is therefore a *within-model ablation*—with vs. without explicit obligation-graph structure—not an architectural comparison across model families. Conclusions about the relative value of graph structure should be confirmed with a second model family before generalisation.

### 4.5 LLM Judge (Flat Pairwise)

Claude Haiku with explicit AGREE / DIVERGE / SILENT definitions, prompted once per pair with both texts. No graph, no retrieval. Full prompt and code at Appendix A.

## 5 Results

Table 1 reports macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for all four methods.

Table 1. Macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for the four baselines ( $n = 101$  pairs).

| Method                               | Macro-F1 [95% CI]          | Acc          | AGREE F1 [CI]              | DIVERGE F1 [CI]            | SILENT F1 [CI]             |
|--------------------------------------|----------------------------|--------------|----------------------------|----------------------------|----------------------------|
| Lexical heuristic <sup>1</sup>       | 0.511 [0.411–0.605]        | 0.541        | 0.718 [0.605–0.809]        | 0.466 [0.320–0.593]        | 0.334 [0.133–0.537]        |
| NLI cross-encoder <sup>2</sup>       | 0.233 [0.152–0.303]        | 0.238        | 0.279 [0.103–0.449]        | 0.383 [0.247–0.506]        | 0.025 [0.000–0.086]        |
| Graph-RAG pair-level <sup>3</sup>    | 0.663 [0.570–0.747]        | 0.673        | 0.754 [0.646–0.851]        | 0.586 [0.433–0.722]        | 0.640 [0.478–0.771]        |
| <b>LLM judge (Haiku)<sup>4</sup></b> | <b>0.830 [0.747–0.908]</b> | <b>0.842</b> | <b>0.905 [0.831–0.965]</b> | <b>0.805 [0.696–0.892]</b> | <b>0.778 [0.609–0.914]</b> |

<sup>1</sup>Nested 5-fold CV, 5 seeds; thresholds tuned on inner folds only. <sup>2</sup>Same CV protocol, thresholds tuned on inner folds. <sup>3</sup>No tunable hyperparameters; bootstrap CI only. <sup>4</sup>claude-haiku-4-5-20251001, default temperature; bootstrap CI only. Estimated cost: ~\$0.10 for all 101 pairs.

Table 2. Per-direction SILENT F1 for all four methods (bootstrap 95% CI,  $n = 101$  pairs). SILENT\_FDA ( $n = 13$ ): FDA is silent, EMA regulates. SILENT\_EMA ( $n = 8$ ): EMA is silent, FDA regulates.

| Method                   | SILENT_FDA F1 [95% CI]     | SILENT_EMA F1 [95% CI]     |
|--------------------------|----------------------------|----------------------------|
| Lexical heuristic        | 0.277 [0.000–0.545]        | 0.393 [0.000–0.706]        |
| NLI cross-encoder        | 0.049 [0.000–0.167]        | 0.000 [0.000–0.000]        |
| Graph-RAG pair-level     | 0.716 [0.516–0.882]        | 0.535 [0.267–0.762]        |
| <b>LLM judge (Haiku)</b> | <b>0.795 [0.588–0.952]</b> | <b>0.738 [0.429–0.952]</b> |

SILENT-FDA: 13 pairs (FDA silent, EMA active). SILENT-EMA: 8 pairs (EMA silent, FDA active). Wide CIs on SILENT-EMA reflect the small stratum ( $n = 8$ ). Bootstrap 1,000 resamples; stratified by 3-class label.

Point estimates increase monotonically lexical  $\rightarrow$  Graph-RAG  $\rightarrow$  LLM judge; bootstrap CIs partially overlap at  $n = 101$ , so we treat this as a ranking trend rather than a significance claim. **Lexical ceiling:** errors share the signature of *high lexical overlap, opposed regulatory stance* (e.g., FDA “modified Mayo score of 5–9” vs. EMA “full Mayo score of 9–12”; same vocabulary, different instrument), requiring semantic understanding of what each requirement asserts.

## 6 Analysis

### 6.1 What the Baseline Hierarchy Reveals

**Observation 1 — SILENT is semantically detectable but invisible to NLI framing.** SILENT F1 = 0.025 under NLI (no native SILENT class) vs. 0.778 under the LLM judge (explicit definition). This contrast reflects a structural constraint—the SILENT class requires a formulation that can express absence, which NLI entailment cannot—rather than a model-capacity effect. A higher-capacity NLI encoder would improve AGREE/DIVERGE F1 but would not resolve the absence-detection gap without task-specific supervision.

**Observation 2 — For Claude Haiku on this dataset, pair-level obligation graphs improve over lexical (+0.15 F1) but trail flat prompting (−0.17 F1); CIs partially overlap at pilot scale.** The graph makes obligation structure explicit, helping distinguish stance differences from similar-sounding text. But it discards full-text context that flat prompting uses to separate *same-requirement* / *different-modal-register* (AGREE) from *same-topic* / *different-threshold* (DIVERGE): gene-01 (FDA 15 years vs. EMA case-by-case) both extract as RECOMMENDED and the graph calls them AGREE; the flat LLM

correctly identifies the threshold conflict from full text [8]. This comparison is a within-model ablation (graph vs. no-graph, both using Haiku), not a cross-model claim.

**Observation 3 — Two distinct problem scopes suggest two distinct architectural targets.** The *pair-level scope* (given pre-aligned sentences, label AGREE/DIVERGE/SILENT) is largely addressed by the flat LLM judge (0.830). The *corpus-level scope* (“is EMA silent anywhere across its corpus?”) requires finding aligned pairs, not just classifying them; a corpus-level policy graph is needed. §7 motivates this architecture as the next contribution.

## 6.2 Source-Stratified Performance Analysis

To assess whether performance reflects source-study-specific patterns rather than genuine task competence, we analyse performance stratified by track and by source concentration.

**Track-level.** On AGREE/DIVERGE classes, the LLM judge scores *lower* on Track A (0.769/0.745) than Track B (0.966/0.923): exposure to source studies does not inflate performance. The macro-F1 gap (Track B 0.630 vs. Track A 0.765) is explained by SILENT support ( $n = 0$  vs.  $n = 21$ ), not memorisation.

**Source-concentration.** 40/101 pairs come from one source (Vieujean et al.). LLM judge on the JCC stratum ( $n = 40$ ): 0.804; on all other pairs ( $n = 61$ ): 0.844. Performance is marginally *higher* outside JCC, ruling out domain-concentration inflation.

## 7 Future Work

**Corpus-level policy graph.** The correct long-term target is corpus-level silent detection: (1) build per-jurisdiction policy graphs indexing all FDA/EMA guidance; (2) align requirement nodes via entity normalisation and embedding similarity; (3) classify aligned pairs as AGREE/DIVERGE/SILENT by obligation-structure comparison; (4) generate natural-language explanations. This architecture subsumes the pair-alignment assumption: alignment becomes a graph-construction step, not a pre-processing input.

**RegDivergence-500.** Four domains are unrepresented in the pilot: CMC, adaptive trial design/estimates, statistical analysis plan requirements, and labelling/SmPC. RegDivergence-500 ( $\approx 500$  pairs) will cover all four via a community-annotation pipeline seeded from a  $\sim 45$ -source FDA/EMA comparison inventory. Annotation guidelines are released with this paper. Multi-model replication, an independent third-annotator validation pass, and a cross-pair generalisation probe against Sino-US-DrugQA [9] are planned; a third jurisdiction (PMDA) will add an  $\approx 40$ -pair FDA $\times$ PMDA subset [10, 11].

## 8 Limitations

**Scale and alignment.** 101 pairs from three source studies limits statistical power to directional observations. Track B co-topicality judgements are single-annotator; Track A inherits source-study alignment methodology. Reproducibility depends on the released exclusion-criteria checklist (`data/alignment_protocol.md`). Oncology, rare diseases, medical devices, and vaccines are absent; RegDivergence-500 (§7) is required before domain-general conclusions.

**SILENT labels.** All 21 SILENT pairs carry `doc_confidence = MODERATE`; treat them as soft labels. The pair-level SILENT task is a proxy for the correct corpus-level formulation (§7).

**Single-model evaluation.** All LLM-dependent components use Claude Haiku. The Graph-RAG vs. LLM comparison is a within-model ablation, not a cross-model comparison. The  $\kappa = 0.542$  LLM consistency check reflects the same model’s label tendencies. Haiku may have seen FDA/EMA guidance during pretraining; the source-stratified analysis (§6.2) provides evidence against inflation, but future work should evaluate on post-cutoff guidance. Multi-model replication is deferred to RegDivergence-500.

**Use norms.** RegDivergence-101 is a pilot benchmark: not sufficient for deployment decisions, not domain-generalizable without held-out evaluation, and not a sole criterion for AI procurement in regulatory-affairs workflows. Conclusions are specific to the FDA/EMA pair until cross-pair generalisation probes are run.

## 9 Conclusion

We introduced cross-jurisdiction regulatory divergence detection and released **RegDivergence-101**, a 101-pair expert-grounded **pilot** benchmark ( $\kappa = 0.85$ , two-track construction protocol, per-direction SILENT annotation) with a four-method baseline hierarchy. Three directional observations at pilot scale: SILENT requires explicit absence-aware formulation (NLI F1 0.03  $\rightarrow$  LLM F1 0.78); for Claude Haiku, pair-level graph structure improves over lexical (+0.15 F1) but trails flat prompting (−0.17 F1); and the pair-level vs. corpus-level scope distinction motivates the corpus-level graph architecture (§7) as the next contribution. RegDivergence-500 will establish the task at production scale.

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This work was conducted independently. Benchmark pairs were derived from publicly available FDA/EMA guidance documents and open-access peer-reviewed comparison studies. LLM API calls used the Anthropic API (Claude Haiku, `claude-haiku-4-5-20251001`).

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## A Prompts and Code

Full prompts (LLM judge system/user, Graph-RAG Stage 1 triplet-extraction, Stage 2 graph-classification), evaluation scripts, and dataset schema are available at the project repository: <https://github.com/chuchugo/cross-jurisdiction-regulatory-divergence>

**LLM judge** (`llm_judge.py`): system prompt defines AGREE/DIVERGE/SILENT with explicit absence framing; user prompt pairs `{fda_text}` and `{ema_text}`. Model: `claude-haiku-4-5-20251001`, temperature 1.0, max tokens 100.

**Graph-RAG** (`graph_rag.py`): Stage 1 extracts a policy node `{subject, obligation_level, requirement, conditions}`; Stage 2 classifies the aligned node pair. Hard rules: MANDATORY  $\leftrightarrow$  PROHIBITED  $\Rightarrow$  DIVERGE (no LLM call); both SILENT  $\Rightarrow$  SILENT. Max tokens: 300.